

# Consolidated Page Manifest and Free-Only UI Integration Plan for BusinessOS AI and an Accounting SaaS

## Executive summary

This report consolidates the **complete route/page inventory**, recommended **free/open-source UI and service stack**, a **page-to-tool implementation map**, hosting/cost tradeoffs, and a **handoff-ready manifest** for two products: **BusinessOS AI** and a **TallyPrime-like multi-tenant Accounting SaaS**. Team size, budget, primary country, payment rails, deployment regions, and AI model vendors are **unspecified**. That matters especially for accounting, because localization, tax, and banking assumptions change page count, workflow depth, and infrastructure choices.

The strongest common foundation is a shared system built on [WCAG 2.2](https://www.w3.org/TR/WCAG22/), [WAI-ARIA Authoring Practices](https://www.w3.org/WAI/ARIA/apg/), [Apple HIG Dark Mode](https://developer.apple.com/design/human-interface-guidelines/dark-mode/), and [Material accessibility guidance](https://m3.material.io/foundations/designing/structure/); visible focus, contrast-safe dark surfaces, status messages announced to assistive tech, robust table/grid semantics, and minimum touch targets of at least 24×24 CSS px by WCAG and typically 48×48 by Material guidance. [1]

The recommended free-first UI stack is: [Next.js App Router](https://nextjs.org/docs/app/) for routing and shells, [Tailwind CSS](https://tailwindcss.com/docs/) for styling, [shadcn/ui](https://ui.shadcn.com/docs) plus [Radix Primitives](https://www.radix-ui.com/primitives/docs/overview/introduction) for accessible building blocks, [TanStack Table](https://tanstack.com/table/latest/docs/introduction) for data-heavy tables, [React Hook Form](https://react-hook-form.com/docs/) with [Zod](https://zod.dev/) for validation, [Tiptap](https://tiptap.dev/docs/editor/getting-started/overview) or [Lexical](https://lexical.dev/docs/intro) for rich text, [Uppy](https://uppy.io/docs/quick-start/) for uploads, [CodeMirror](https://codemirror.net/docs/) for code-like editors, [Lucide Icons](https://lucide.dev/guide/) for icons, and [axe-core](https://github.com/dequelabs/axe-core) plus [Playwright accessibility testing](https://playwright.dev/docs/accessibility-testing) for QA. Those choices are strongly supported by official docs and align with a dark, keyboard-first product shell. [2]

For shared services, the best free-first pattern is pooled frontend/UI hosting on [Cloudflare Pages](https://developers.cloudflare.com/pages/) or [Vercel](https://vercel.com/)

Hobby<https://vercel.com/docs/plans/hobby>, relational data on [PostgreSQL](https://www.postgresql.org/docs/current/ddl-rowsecurity.html)<https://www.postgresql.org/docs/current/ddl-rowsecurity.html> via [Neon pricing](https://neon.com/pricing)<https://neon.com/pricing> or [Supabase pricing](https://supabase.com/pricing)<https://supabase.com/pricing>, auth via [Auth.js](https://authjs.dev/getting-started)<https://authjs.dev/getting-started> or [Keycloak docs](https://www.keycloak.org/documentation)<https://www.keycloak.org/documentation>, jobs via [pg-boss](https://timgit.github.io/pg-boss/)<https://timgit.github.io/pg-boss/> first and [BullMQ](https://docs.bullmq.io/)<https://docs.bullmq.io/> when Redis is justified, storage via [SeaweedFS](https://seaweedfs.com/)<https://seaweedfs.com/> or [MinIO](https://docs.min.io/aistor/)<https://docs.min.io/aistor/> for self-hosting and [Cloudflare R2](https://developers.cloudflare.com/r2/pricing/)<https://developers.cloudflare.com/r2/pricing/> for managed object storage, analytics via [PostHog pricing](https://posthog.com/pricing)<https://posthog.com/pricing> or [Matomo free software](https://matomo.org/free-software/)<https://matomo.org/free-software/>, BI via [Metabase OSS](https://www.metabase.com/start/oss/)<https://www.metabase.com/start/oss/>, and optional media via [tusd](https://tus.github.io/tusd/)<https://tus.github.io/tusd/>, [FFmpeg](https://ffmpeg.org/documentation.html)<https://ffmpeg.org/documentation.html>, [Shaka Packager docs](https://shaka-project.github.io/shaka-packager/html/documentation.html)<https://shaka-project.github.io/shaka-packager/html/documentation.html>, [Video.js](https://videojs.com/)<https://videojs.com/>, and [OvenMediaEngine](https://docs.ovenmediaengine.com/)<https://docs.ovenmediaengine.com/> or [MediaMTX](https://mediamtx.org/)<https://mediamtx.org/>. [3]

## Shared architecture and component-to-tool map

The platform should use one shared design system and two product shells: a **workflow shell** for BusinessOS AI and a **ledger-grade shell** for Accounting. The former privileges dashboards, editors, timelines, and AI context panels; the latter privileges dense tables, line-entry forms, reconciliation panes, posting reviews, and immutable-state cues. [4]

flowchart LR

DS[Shared design system<br/>tokens, accessibility, templates]

UI1[BusinessOS AI shell]

UI2[Accounting SaaS shell]

DS --> UI1

DS --> UI2

UI1 --> FE[Next.js + Tailwind + shadcn/ui + Radix]

UI2 --> FE

FE --> AUTH[Auth.js or Keycloak]

FE --> DB[Postgres via Neon or Supabase]

FE --> JOBS[pg-boss first / BullMQ when Redis exists]

FE --> STORE[SeaweedFS or MinIO / R2 managed]

FE --> ANALYTICS[PostHog or Matomo + Metabase]

FE --> MEDIA[Optional: tusd + FFmpeg + Shaka + Video.js + OME/MediaMTX]

## Compact component-to-tool mapping

| Component / page surface             | Preferred OSS tools                                                                                                                                                                                                                                                          | Strong alternatives                                | Best-fit notes                                                                                   |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------|
| App shell, route layouts, docs pages | <a href="#">Next.js App Router</a> <a href="https://nextjs.org/docs/app">https://nextjs.org/docs/app</a> + <a href="#">Tailwind CSS</a> <a href="https://tailwindcss.com/docs">https://tailwindcss.com/docs</a>                                                              | unspecified SSR alternatives if team stack differs | Use for all shells, auth, marketing, docs, help.                                                 |
| Accessible primitives                | <a href="#">shadcn/ui</a> <a href="https://ui.shadcn.com/docs">https://ui.shadcn.com/docs</a> + <a href="#">Radix Primitives</a> <a href="https://www.radix-ui.com/primitives/docs/overview/introduction">https://www.radix-ui.com/primitives/docs/overview/introduction</a> | unspecified if design system is fully in-house     | Best for dialogs, dropdowns, tabs, popovers, drawers.                                            |
| Tables / grids / tree tables         | <a href="#">TanStack Table</a> <a href="https://tanstack.com/table/latest/docs/introduction">https://tanstack.com/table/latest/docs/introduction</a>                                                                                                                         | unspecified grid alternatives                      | Use read-heavy tables in both products; editable journal grids still need custom keyboard model. |
| Form                                 | <a href="#">React Hook Form</a> <a href="https://react-hook-form.com">https://react-hook-form.com</a>                                                                                                                                                                        | unspecified schema libs                            | Strong                                                                                           |

| Component / page surface               | Preferred OSS tools                                                                              | Strong alternatives                                  | Best-fit notes                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------|
| Forms / validation                     | <a href="https://hook-form.com/docs">hook-form.com/docs</a> + <a href="https://zod.dev/">Zod</a> |                                                      | for dynamic forms, wizards, invoice/journal entry.                        |
| Rich text / proposal / article editing | <a href="https://tiptap.dev/docs/editor/getting-started/overview">Tiptap</a>                     | <a href="https://lexical.dev/docs/intro">Lexical</a> | Tiptap for Notion-like content blocks; Lexical for leaner custom editors. |
| Code blocks / JSON / rules editors     | <a href="https://codemirror.net/docs/">CodeMirror</a>                                            | unspecified                                          | Use for API payload preview, workflow expressions, template JSON.         |
| File manager / uploads                 | <a href="https://uppy.io/docs/quick-start/">Uppy</a>                                             | unspecified                                          | Use Dashboard UI for assets, attach                                       |

| Component / page surface | Preferred OSS tools                                                                                                                       | Strong alternatives                                           | Best-fit notes                                                                                      |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Charts / analytics       | <a href="https://recharts.org/">Recharts</a>                                                                                              | unspecified charting options                                  | ments, statements, invoices.<br>Good default for KPI + line/bar charts; pair with tabular fallback. |
| Icons                    | <a href="https://lucide.dev/guide/">Lucide Icons</a>                                                                                      | unspecified                                                   | Consistent monochrome iconography for dark theme.                                                   |
| Accessibility QA         | <a href="https://github.com/dequelabs/axe-core">axe-core</a> + <a href="https://playwright.dev/docs/accessibility-testing">Playwright</a> | manual SR testing remains required                            | Automated checks plus scripted keyboard flows.                                                      |
| Auth / tenant session    | <a href="https://authjs.dev/getting-started">Auth.js</a>                                                                                  | <a href="https://www.keycloak.org/documentation">Keycloak</a> | Auth.js for MVP, Keycloak when                                                                      |

| Component / page surface          | Preferred OSS tools                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Strong alternatives                                                                                                                                                                                                                                                                                    | Best-fit notes                                             |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| on flows                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                        | SSO/federation matters .                                   |
| Data / jobs / storage / analytics | <a href="https://www.postgresql.org/docs/current/ddl-rowsecurity.html">?url?PostgreSQL RLS?https://www.postgresql.org/docs/current/ddl-rowsecurity.html?</a> +<br><a href="https://neon.com/pricing">?url?Neon?https://neon.com/pricing?</a> or<br><a href="https://supabase.com/pricing">?url?Supabase?https://supabase.com/pricing?</a> + <a href="https://timgit.github.io/pg-boss/">?url?pg-boss?https://timgit.github.io/pg-boss/?</a> +<br><a href="https://seaweedfs.com/">?url?SeaweedFS?https://seaweedfs.com/?</a> or<br><a href="https://docs.min.io/aistor/">?url?MinIO?https://docs.min.io/aistor/?</a> +<br><a href="https://posthog.com/pricing">?url?PostHog?https://posthog.com/pricing?</a> +<br><a href="https://www.metabase.com/start/oss/">?url?Metabase?https://www.metabase.com/start/oss/?</a> | <a href="https://docs.bullmq.io/">?url?BullMQ?https://docs.bullmq.io/?</a> ,<br><a href="https://matomo.org/free-software/">?url?Matomo?https://matomo.org/free-software/?</a> ,<br><a href="https://developers.cloudflare.com/r2/pricing/">?url?R2?https://developers.cloudflare.com/r2/pricing/?</a> | Shared backend service bundle.                             |
| Accounting workflow references    | <a href="https://docs.frappe.io/erp-next/chart-of-accounts">?url?ERPNext?https://docs.frappe.io/erp-next/chart-of-accounts?</a> , <a href="https://akaunting.com/hc/docs/double-entry-accounting">?url?Akaunting docs?https://akaunting.com/hc/docs/double-entry-accounting/?</a> , <a href="https://www.odoo.com/documentation/19.0/applications/finance/accounting.html">?url?Odoo docs?https://www.odoo.com/documentation/19.0/applications/finance/accounting.html?</a> , <a href="https://book.ledgersmb.org/dev/full-book/">?url?LedgerSMB docs?https://book.ledgersmb.org/dev/full-book/?</a>                                                                                                                                                                                                                    | n/a                                                                                                                                                                                                                                                                                                    | Use as reference implementations, not as the permanent UI. |
| Optional media/video surfaces     | <a href="https://tus.github.io/tusd/">?url?tusd?https://tus.github.io/tusd/?</a> +<br><a href="https://ffmpeg.org/documentation.html">?url?FFmpeg?https://ffmpeg.org/documentation.html?</a> + <a href="https://shaka-project.github.io/shaka-packager/html/documentation.html">?url?Shaka Packager?https://shaka-project.github.io/shaka-packager/html/documentation.html?</a> +<br><a href="https://videojs.com/">?url?Video.js?https://videojs.com/?</a>                                                                                                                                                                                                                                                                                                                                                             | <a href="https://shaka-player-demo.appspot.com/docs/api/tutorial-welcome.html">?url?Shaka Player?https://shaka-player-demo.appspot.com/docs/api/tutorial-welcome.html?</a> ,<br><a href="https://docs.ovenmediaengine.com/">?url?OvenMediaEngine?https://docs.ovenmediaengine.com/?</a> ,              | Only if either product later needs media hosting           |

Component /  
page surface

| Preferred OSS tools | Strong alternatives                                                                                    | Best-fit notes |
|---------------------|--------------------------------------------------------------------------------------------------------|----------------|
|                     | <a href="https://mediaMTX.com">MediaMTX</a><br><a href="https://mediaMTX.com">https://mediaMTX.com</a> | /playback.     |

## BusinessOS AI consolidated page manifest

**Baseline for all routes:** T-MKT marketing, T-AUTH auth split, T-WIZ wizard, T-SHELL app shell, T-INDEX list/index, T-DETAIL detail split, T-EDITOR canvas/editor, T-REPORT reporting, T-DOCS docs/help, T-ADMIN admin. Default toolkit is the shared stack above; rows below list only route-specific emphasis.

| Route / route group                                          | Purpose; personas         | Required data / props                                 | Key components and OSS fit                                                                                                                                           | States                               | Responsive / accessibility / micro-interactions                                         |
|--------------------------------------------------------------|---------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------|
| /, /features, /templates, /pricing, /integrations, /security | Acquisition; OWN/MKT/PM   | hero copy, feature cards, plan matrix, template cards | marketing cards, comparison tables, FAQ; baseline stack                                                                                                              | load, no-results, form success/error | 12→4 col collapse; accordions on mobile; CTA and pricing toggles announce state changes |
| /docs/*, /help/*, /status, /changelog                        | Docs, trust, support; all | nav tree, article body, TOC, incident feed            | docs shell, search, code blocks via <a href="https://codemirror.net/docs/">CodeMirror</a><br><a href="https://codemirror.net/docs/">https://codemirror.net/docs/</a> | 404, empty search, degraded status   | sticky TOC must not obscure focus; status colors always                                 |

| Route / route group                                | Purpose; personas         | Required data / props                                   | Key components and OSS fit           | States                                   | Responsive / accessibility / micro-interactions                  |
|----------------------------------------------------|---------------------------|---------------------------------------------------------|--------------------------------------|------------------------------------------|------------------------------------------------------------------|
| /login, /signup, /forgot-password, /reset-password | Auth; all                 | providers, email, password, SSO slug, token             | auth forms via RHF+Zod               | inline error, mail-sent, expired token   | text-paired password-manager-friendly; live status announcements |
| /workspace/select, /onboarding/*                   | Tenant start; OWN         | workspaces, invite token, goals, brand, starter objects | wizard, stepper, summary cards       | save-draft, validation, complete         | one primary action/step; step state announced                    |
| /app                                               | Dashboard; OWN/SAL/PM/FIN | KPIs, tasks, deals, invoices, AI suggestions            | KPI cards, charts, timeline, AI dock | first-run empty, partial error, refresh  | cards stack on mobile; persistent quick-create                   |
| /app/sites                                         | Site index; MKT           | sites, status, domains, traffic                         | index table/cards, filters           | empty, publishing, pending, deploy error | card mode on mobile                                              |
| /app/sites/:site                                   | Site                      | page tree,                                              | detail split, tabs, activity         | no                                       | tabs                                                             |



| Route / route group                                     | Purpose; personas                | Required data / props                       | Key components and OSS fit                                                                                                                                                                                                                                                | States                                              | Responsive / accessibility / micro-interactions                       |
|---------------------------------------------------------|----------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------|
| teId                                                    | overview; MKT/OWN                | traffic, forms, release summary             |                                                                                                                                                                                                                                                                           | traffic, domain verify failed                       | manual activation                                                     |
| /app/sites/:siteId/editor                               | Website/page editing; MKT        | page JSON, components, assets, theme tokens | editor canvas; <a href="https://tiptap.dev/docs/editor/getting-started/overview">urlTiptaphttps://tiptap.dev/docs/editor/getting-started/overview</a> for copy fragments, <a href="https://uppy.io/docs/dashboard/">urlUppyhttps://uppy.io/docs/dashboard/</a> for assets | dirty, block error, publish under way               | one drawer at a time on tablet/mobile; keyboard reorder path required |
| /app/sites/:siteId/settings, /app/sites/:siteId/publish | SEO, domains, releases, rollback | SEO props, DNS state, SSL state, releases   | forms, timelines, confirm modals                                                                                                                                                                                                                                          | duplicate slug, DNS pending, rollback success/error | destructive actions use confirm/review pattern                        |
| /app/forms                                              | Submission inbox; MKT/SAL/SUP    | schema, submissions, spam flags, routing    | tables, filters, detail drawer                                                                                                                                                                                                                                            | empty, delivery fail                                | sticky selection bar; keyboard row open                               |
| /app/crm/contacts, /app/crm/contacts/:id                | Contacts and detail; SAL/PM      | contacts, accounts, tags, timeline, linked  | tables, profile cards, timeline                                                                                                                                                                                                                                           | import error, no activity ,                         | timeline is list semantics; note compos                               |

| Route / route group                                     | Purpose; personas                      | Required data / props                             | Key components and OSS fit                          | States                                       | Responsive / accessibility / micro-interactions                         |
|---------------------------------------------------------|----------------------------------------|---------------------------------------------------|-----------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------|
|                                                         |                                        | docs                                              |                                                     | archived                                     | er shortcut                                                             |
| /app/crm/deals, /app/crm/deals/:id                      | Pipeline and deal conversion; SAL/FIN  | stages, values, linked proposals /invoices        | kanban + detail split                               | empty pipeline, drag blocked, won/lost       | drag has keyboard equivalent                                            |
| /app/projects, /app/tasks, /app/files                   | Delivery work; PM/OWN                  | project health, tasks, files, permissions         | index, kanban/list, file manager via Uppy + storage | empty, upload in progress, permission denied | files grid collapse to list on mobile                                   |
| /app/proposals, /app/proposals/:id/editor               | Proposal workspace; SAL/FIN            | sections, pricing blocks, approvals               | editor canvas + rich text via Tiptap/Lexical        | draft, sent, expired, approval blocked       | send flow has review step                                               |
| /app/invoices, /app/invoices/:id, /app/portal/:clientId | Invoice ops and client portal; FIN/CLT | invoices, line items, payment status, shared docs | tables, detail split, portal shell                  | overdue, payment fail, invite expired        | money is right-aligned and text-labeled; portal strips internal actions |

| Route / route group                            | Purpose; personas                          | Required data / props                            | Key components and OSS fit                  | States                                    | Responsive / accessibility / micro-interactions                    |
|------------------------------------------------|--------------------------------------------|--------------------------------------------------|---------------------------------------------|-------------------------------------------|--------------------------------------------------------------------|
| /app/automations, /app/automations/:id/builder | Rule list and builder; OWN/DEV             | triggers, steps, run history, schemas            | index + builder; CodeMirror for expressions | invalid rule, test pass/fail              | node graph must also have step-list mode                           |
| /app/reports, /app/ai                          | Analytics and copilot hub; OWN/PM/FIN      | date ranges, report defs, AI threads and context | charts, tables, AI panel                    | no data, model unavailable, needs confirm | charts need text fallbacks; AI actions confirm before side effects |
| /app/settings/*, /app/developer/*              | org settings, branding, API keys, webhooks | team list, scopes, secrets, theme tokens         | forms, tables, code blocks                  | revoke key, contrast warning              | secrets masked by default; contrast checker badge                  |
| /admin/*                                       | super-admin ops                            | tenants, plans, health, audit logs               | admin console, dense tables, diff drawers   | degraded service, export queue            | typed confirmation for high-risk actions                           |

BusinessOS AI's route set is UI-broad but technically cohesive: editor-heavy pages lean on Tiptap/Lexical/Uppy, list-heavy pages lean on TanStack Table, and everything else reuses the common shell. That keeps the design system compact even though the product surface is large. [5]

## Accounting SaaS consolidated page manifest

The accounting product uses the same templates except one special page type: T-RECON, a reconciliation workspace with split panes and high-density keyboard behavior.

| Route / route group                               | Purpose; personas                                   | Required data / props                                  | Key components and OSS fit              | States                                     | Responsive / accessibility / micro-interactions                     |
|---------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|-----------------------------------------|--------------------------------------------|---------------------------------------------------------------------|
| /, /features, /pricing, /localizations, /security | Acquisition and compliance positioning; OWN/ACC/TAX | features, country packs, report previews, plans        | marketing shell, comparison tables      | load, unsupported locale                   | legal copy readable on mobile; plain-language disclaimers           |
| /docs/*, /help/*, /status, /changelog             | Help, glossary, release changes; ACC/BK/DEV         | nav tree, articles, glossary, change notes             | docs shell, searchable KB, code blocks  | 404, no results                            | glossary terms exposed through focusable disclosure, not hover only |
| /login, /signup, /company/select, /onboarding/*   | Auth and company setup; OWN/ACC                     | company, fiscal year, currency, tax mode, COA template | auth split + wizard                     | validation, country-pack warning, complete | step order is explicit; numeric/date hints always visible           |
| /app                                              | Accounting dashboard; OWN/ACC/BK                    | cash, AR/AP aging, tax due, reconciliation queue       | KPI cards, alert stack, charts + tables | first-run, stale feed, tax warning         | top section answers “what needs attention now”                      |
| /app/accounts, /app/accounts/:id                  | Chart of accounts and account detail; ACC/AUD       | account code, type, parent, balances, postings         | tree-table + detail split               | empty COA, locked template                 | tree controls need robust keyboard expansion/collapse               |

| Route / route group                                                                     | Purpose;<br>personas                             | Required<br>data /<br>props                                              | Key<br>components<br>and OSS fit                             | States                                                                   | Responsive /<br>accessibility /<br>micro-<br>interactions             |
|-----------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------|
| /app/journals                                                                           | Journal list;<br>ACC/AUD                         | draft/posted<br>entries,<br>source<br>docs,<br>periods                   | dense table,<br>filters,<br>saved views                      | none,<br>import fail                                                     | posted-state<br>badge always<br>visible                               |
| /app/journals/new<br>,<br>/app/journals/:id<br>/review                                  | Draft,<br>review,<br>post<br>journal;<br>ACC/AUD | lines,<br>dimensio<br>ns,<br>debit/cre<br>dit totals,<br>attachme<br>nts | editable line<br>grid, totals<br>footer,<br>confirm<br>modal | unbalanc<br>ed,<br>blocked<br>period,<br>posted                          | tab-first entry;<br>post uses<br>review-<br>confirm-<br>reverse       |
| /app/customers,<br>/app/vendors                                                         | Master<br>data;<br>AR/AP/ACC                     | identifiers<br>, terms,<br>balances,<br>tax IDs                          | index<br>tables/form<br>s                                    | import<br>fail,<br>duplicate<br>warning                                  | all status<br>badges text-<br>labeled                                 |
| /app/sales-<br>invoices,<br>/app/sales-<br>invoices/new,<br>/app/sales-<br>invoices/:id | AR<br>invoicing;<br>AR/ACC                       | customer<br>, lines,<br>tax, due<br>dates,<br>payments                   | forms,<br>totals rail,<br>detail split                       | draft, tax<br>warning,<br>partial<br>paid                                | send/post<br>step<br>summarizes<br>totals and tax<br>before<br>commit |
| /app/purchase-<br>bills,<br>/app/purchase-<br>bills/new                                 | AP bills;<br>AP/ACC                              | vendor,<br>lines,<br>expense<br>accounts,<br>tax                         | forms,<br>attachment<br>s via Uppy                           | draft,<br>duplicate<br>suspicion,<br>OCR<br>pending if<br>later<br>added | upload never<br>blocks<br>keyboard path                               |
| /app/payments,<br>/app/payments/new                                                     | receipts,<br>payments,<br>allocations<br>; AR/AP | counterp<br>arty,<br>account,<br>amount,<br>allocatio<br>ns              | index +<br>allocation<br>grid                                | over-<br>allocation<br>, partial<br>allocation<br>, reversed             | numeric input<br>optimized for<br>mobile<br>keypad                    |
| /app/bank-<br>accounts                                                                  | linked bank<br>and                               | masked<br>numbers,                                                       | cards/table                                                  | disconnec<br>ted, stale                                                  | masked by<br>default; reveal                                          |

| Route / route group                 | Purpose; personas                                         | Required data / props                               | Key components and OSS fit                                            | States                               | Responsive / accessibility / micro-interactions                            |
|-------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------|
|                                     | manual accounts; ACC                                      | status, last sync                                   |                                                                       | sync                                 | is deliberate                                                              |
| /app/reconciliation                 | statement matching workspace; ACC/BK                      | lines, suggestions, confidence, match targets       | T-RECON, split list/match/action panes; table + detail panel          | all reconciled, conflict, offline    | keyboard shortcuts are mandatory; focus must remain anchored between panes |
| /app/tax/settings, /app/tax/returns | tax configuration and filing prep; TAX/ACC                | registrations, codes, rates, periods, return status | forms, tables, timelines                                              | overlap warning, draft return, filed | effective-date changes need diff/timeline cues                             |
| /app/reports, /app/reports/*        | P&L, balance sheet, GL, trial balance, aging; OWN/ACC/AUD | range, comparison, grouping, export props           | report shell, dense tables, optional charts                           | no data, export pending              | textual totals before charts; column visibility keyboard-operable          |
| /app/period-close                   | close/reopen periods; OWN/ACC/AUD                         | checklist, unresolved items, warnings               | detail split, checklist, confirmations                                | blocked, closed, reopen requested    | legally/financially sensitive confirmation pattern                         |
| /app/imports                        | opening balance and migration jobs; ACC/BK/DEV            | mappings, job logs, diff previews                   | tables, drawers, mapping grid via CodeMirror for transforms if needed | parsing, validation fail, applied    | preserve mapping context on small screens                                  |

| Route / route group               | Purpose; personas                            | Required data / props                                | Key components and OSS fit       | States                                 | Responsive / accessibility / micro-interactions |
|-----------------------------------|----------------------------------------------|------------------------------------------------------|----------------------------------|----------------------------------------|-------------------------------------------------|
| /app/settings/*, /app/developer/* | company settings, users/roles , API/webhooks | company defaults, approvals , scopes                 | forms, tables, code blocks       | save error, revoked key                | approval-role changes need summary modal        |
| /admin/*                          | tenant ops, localization packs, audit        | tenants, queue health, pack versions, audit evidence | admin console, tables, timelines | degraded filing service, suspended org | typed confirmation for high-risk actions        |

The accounting surface is route-narrower than BusinessOS AI, but individual pages are denser and more stateful. The most complex UI is reconciliation, followed by journal review/post and period close. Official docs from Akaunting, Odoo, ERPNext, and LedgerSMB all point to the same persistent workflow spine: chart of accounts, journal entry, tax, bank reconciliation, and financial reporting. [6]

## Hosting, cost tradeoffs, and free-only implementation roadmap

### Hosting and free-tier tradeoffs

| Layer            | Best free-first option                                                                                                                                                         | What official docs say                                                                                              | Where free breaks                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Frontend hosting | <a href="https://developers.cloudflare.com/pages/">?url?Cloudflare Pages</a> <a href="https://developers.cloudflare.com/pages/">?https://developers.cloudflare.com/pages/?</a> | Pages free plan documents 500 deploys/month; Workers free plan shows 100,000 requests/day and 10 ms CPU/invoke. [7] | Heavy server logic, preview environments, or worker-heavy APIs |
| Alternate        | <a href="https://vercel.com/">?url?Vercel</a>                                                                                                                                  | Hobby is                                                                                                            | Team                                                           |

| Layer                      | Best free-first option                                                                                                                                                                                                                                                                | What official docs say                                                                       | Where free breaks                                              |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| frontend                   | Hobby <sup>[2]</sup> <a href="https://vercel.com/docs/plans/hobby">https://vercel.com/docs/plans/hobby</a> <sup>[2]</sup>                                                                                                                                                             | free and aimed at personal/sm all-scale apps. <a href="#">[8]</a>                            | workflows, higher function usage                               |
| Postgres                   | <a href="#">url</a> <sup>[2]</sup> Neon Free <sup>[2]</sup> <a href="https://neon.com/pricing">https://neon.com/pricing</a> <sup>[2]</sup>                                                                                                                                            | 100 projects, 100 CU-hours/project, 0.5 GB storage/project, 5 GB egress. <a href="#">[9]</a> | Heavy analytics, large tenants, sustained compute              |
| All-in-one DB/auth/storage | <a href="#">url</a> <sup>[2]</sup> Supabase Free <sup>[2]</sup> <a href="https://supabase.com/pricing">https://supabase.com/pricing</a> <sup>[2]</sup>                                                                                                                                | 50,000 MAUs, 500 MB DB, 1 GB storage, 5 GB egress. <a href="#">[10]</a>                      | Storage-heavy file workflows, SSO, production BI               |
| Object storage             | <a href="#">url</a> <sup>[2]</sup> Cloudflare R2 <sup>[2]</sup> <a href="https://developers.cloudflare.com/r2/pricing/">https://developers.cloudflare.com/r2/pricing/</a> <sup>[2]</sup>                                                                                              | 10 GB-month free + 1M Class A and 10M Class B operations. <a href="#">[11]</a>               | High read volume, media-heavy apps, analytics exports          |
| Self-host storage          | <a href="#">url</a> <sup>[2]</sup> SeaweedFS <sup>[2]</sup> <a href="https://seaweedfs.com/">https://seaweedfs.com/</a> <sup>[2]</sup> / <a href="#">url</a> <sup>[2]</sup> MinIO <sup>[2]</sup> <a href="https://docs.min.io/aistor/">https://docs.min.io/aistor/</a> <sup>[2]</sup> | OSS/self-host; no vendor bill, but you own ops. <a href="#">[12]</a>                         | You pay in servers, backups, and ops time instead of SaaS fees |
| Analytics                  | <a href="#">url</a> <sup>[2]</sup> PostHog <sup>[2]</sup> <a href="https://posthog.com/pricing">https://posthog.com/pricing</a> <sup>[2]</sup>                                                                                                                                        | Official pricing advertises 1M events/mo and 5,000 recordings/                               | High-volume product analytics, long retention                  |



| Layer               | Best free-first option                                                                        | What official docs say                                              | Where free breaks                                             |
|---------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------|
| Transactional email | <a href="https://resend.com/pricing">Resend</a>                                               | 3,000 emails/month and 100/day on free. <a href="#">[14]</a>        | Production notification volume, multi-domain setups           |
| Video optional      | OSS stack or <a href="https://developers.cloudflare.com/stream/pricing">Cloudflare Stream</a> | Stream prices by minutes stored and delivered. <a href="#">[15]</a> | Free-only is usually insufficient once playback traffic grows |

### Prioritized free-only implementation roadmap

| Window     | Pages / modules                                            | OSS tools                                            | Deliverable                                       | Effort  |
|------------|------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------|---------|
| Days 1–10  | shared shells, auth, marketing, docs/help                  | Next.js, Tailwind, shadcn/ui, Radix, Auth.js, Lucide | design-system shell + public pages + auth routes  | 5–6 pw  |
| Days 11–20 | BusinessOS dashboard, sites index/editor, CRM index/detail | TanStack Table, RHF+Zod, Tiptap, Uppy, pg-boss       | first interactive BusinessOS workspace            | 7–8 pw  |
| Days 11–20 | Accounting dashboard, COA, journals list/new/review        | TanStack Table, RHF+Zod, CodeMirror, pg-boss         | first accounting workspace with posting review UI | 8–9 pw  |
| Days 21–30 | BusinessOS proposals, invoices, portal                     | Tiptap, RHF+Zod, Postgres, email                     | commercial ops surfaces                           | 6–7 pw  |
| Days 21–30 | Accounting invoices, bills, payments                       | RHF+Zod, dense line-grid patterns                    | AR/AP surfaces                                    | 7–8 pw  |
| Days 31–40 | BusinessOS automations, reports, AI panel                  | CodeMirror, Recharts, PostHog, Metabase              | workflows + analytics + copilot shell             | 6–7 pw  |
| Days 31–40 | Accounting reconciliation, tax settings, returns           | TanStack + custom keyboard model, charts/tables      | high-density accounting flows                     | 8–10 pw |

| Window        | Pages / modules                                                       | OSS tools                               | Deliverable                          | Effort    |
|---------------|-----------------------------------------------------------------------|-----------------------------------------|--------------------------------------|-----------|
| Days<br>41–50 | settings, developer<br>portals, admin<br>consoles                     | CodeMirror, tables,<br>audit views      | platform/admin/dev<br>routes         | 5–6<br>pw |
| Days<br>51–60 | accessibility<br>hardening, QA,<br>sample data,<br>generator manifest | axe-core, Playwright,<br>PostHog/Matomo | full annotated manifest<br>+ QA pass | 5–6<br>pw |

gantt

title 60-day UI + integration work  
dateFormat YYYY-MM-DD

section Shared

Shells, auth, marketing, docs :a1, 2026-05-09, 10d  
Design system and accessibility :a2, 2026-05-12, 14d

section BusinessOS AI

Dashboard, sites, CRM :b1, 2026-05-19, 12d  
Proposals, invoices, portal :b2, 2026-05-31, 10d  
Automations, reports, AI :b3, 2026-06-10, 10d

section Accounting SaaS

COA, journals, posting review :c1, 2026-05-19, 12d  
Invoices, bills, payments :c2, 2026-05-31, 10d  
Reconciliation, tax, reports :c3, 2026-06-10, 12d

section Platform

Settings, developer, admin :d1, 2026-06-20, 8d  
QA, sample data, handoff manifest :d2, 2026-06-26, 8d

## Handoff-ready manifest for an AI UI generator

**Template IDs:** T-MKT, T-AUTH, T-WIZ, T-SHELL, T-INDEX, T-DETAIL, T-EDITOR, T-REPORT, T-RECON, T-DOCS, T-ADMIN.

## BusinessOS AI CSV-style manifest

| route    | templa<br>te | primary<br>compone<br>nts                                   | sample data keys | mock API<br>endpoints   |
|----------|--------------|-------------------------------------------------------------|------------------|-------------------------|
| /        | T-MKT        | hero, kpi-<br>card,<br>feature-<br>grid, faq,<br>footer-cta | marketing.home.* | GET<br>/api/public/home |
| /pricing | T-MKT        | pricing-                                                    | plans.*          | GET                     |

| route                         | templa<br>te     | primary<br>compon<br>e<br>nts                                       | sample data keys | mock API<br>endpoints                |
|-------------------------------|------------------|---------------------------------------------------------------------|------------------|--------------------------------------|
|                               |                  | cards,<br>compare-<br>table, faq                                    |                  | /api/public/plans                    |
| /login                        | T-<br>AUTH       | auth-<br>form,<br>oauth-<br>list,<br>status-<br>banner              | auth.providers.* | POST /api/auth/*                     |
| /workspace/select             | T-WIZ            | workspac<br>e-cards,<br>create-<br>cta                              | workspaces.*     | GET<br>/api/workspaces               |
| /onboarding/*                 | T-WIZ            | stepper,<br>form-<br>sections,<br>summary<br>-card                  | onboarding.*     | GET/POST<br>/api/onboarding/<br>*    |
| /app                          | T-<br>SHELL      | kpi-card,<br>charts,<br>timeline,<br>quick-<br>actions,<br>ai-panel | dashboard.*      | GET<br>/api/dashboard                |
| /app/sites                    | T-<br>INDEX      | data-<br>table,<br>filters,<br>empty-<br>state                      | sites.list.*     | GET /api/sites                       |
| /app/sites/:siteId/edi<br>tor | T-<br>EDITO<br>R | left-<br>library,<br>canvas,<br>inspector,<br>asset-<br>manager     | site.editor.*    | GET/PUT<br>/api/sites/:id/e<br>ditor |
| /app/crm/contacts             | T-<br>INDEX      | data-<br>table,<br>filter-bar,<br>bulk-                             | crm.contacts.*   | GET<br>/api/contacts                 |

| route                            | templa<br>te     | primary<br>compon<br>ents                                                   | sample data keys         | mock API<br>endpoints               |
|----------------------------------|------------------|-----------------------------------------------------------------------------|--------------------------|-------------------------------------|
|                                  |                  | actions                                                                     |                          |                                     |
| /app/crm/deals/:id               | T-<br>DETAIL     | summary<br>-rail,<br>timeline,<br>linked-<br>docs,<br>actions               | crm.deal.*               | GET<br>/api/deals/:id               |
| /app/proposals/:id/edi<br>tor    | T-<br>EDITO<br>R | rich-text,<br>pricing-<br>blocks,<br>comment<br>s,<br>approvals             | proposal.editor.*        | GET/PUT<br>/api/proposals/:<br>id   |
| /app/invoices/:id                | T-<br>DETAIL     | invoice-<br>preview,<br>status-<br>badges,<br>timeline,<br>payment-<br>card | invoice.detail.*         | GET<br>/api/invoices/:i<br>d        |
| /app/automations/:id/b<br>uilder | T-<br>EDITO<br>R | rule-list,<br>expressio<br>n-editor,<br>run-log                             | automation.build<br>er.* | GET/PUT<br>/api/automations<br>/:id |
| /app/reports                     | T-<br>REPOR<br>T | report-<br>cards,<br>charts,<br>tables,<br>date-<br>range                   | reports.*                | GET<br>/api/reports/*               |
| /app/ai                          | T-<br>SHELL      | ai-panel,<br>context-<br>picker,<br>source-<br>chips,<br>confirm-<br>drawer | ai.*                     | GET/POST<br>/api/ai/*               |
| /app/settings/*                  | T-<br>DETAIL     | forms,<br>tables,                                                           | settings.*               | GET/PUT<br>/api/settings/*          |

| route            | templa<br>te | primary<br>compone<br>nts                            | sample data keys | mock API<br>endpoints   |
|------------------|--------------|------------------------------------------------------|------------------|-------------------------|
|                  |              | secrets,<br>theme-<br>preview                        |                  |                         |
| /app/developer/* | T-<br>DOCS   | endpoint-<br>cards,<br>code-<br>blocks,<br>key-table | developer.*      | GET<br>/api/developer/* |
| /admin/*         | T-<br>ADMIN  | tables,<br>audit-diff,<br>system-<br>kpis            | admin.*          | GET /api/admin/*        |

### Accounting SaaS CSV-style manifest

| route          | templ<br>ate | primary<br>compone<br>nts                                            | sample data keys            | mock API endpoints                           |
|----------------|--------------|----------------------------------------------------------------------|-----------------------------|----------------------------------------------|
| /              | T-MKT        | hero,<br>feature-<br>grid,<br>localizati<br>on-<br>cards,<br>pricing | marketing.account<br>ing.*  | GET<br>/api/public/accounti<br>ng-home       |
| /localizations | T-MKT        | country-<br>cards,<br>support-<br>matrix,<br>faq                     | localizations.*             | GET<br>/api/public/localiza<br>tions         |
| /login         | T-<br>AUTH   | auth-<br>form,<br>org-slug,<br>status-<br>banner                     | auth.*                      | POST /api/auth/*                             |
| /onboarding/*  | T-WIZ        | fiscal-<br>wizard,<br>coa-<br>template<br>-picker,                   | onboarding.account<br>ing.* | GET/POST<br>/api/onboarding/acco<br>unting/* |

| route                    | template | primary components                                             | sample data keys     | mock API endpoints                     |
|--------------------------|----------|----------------------------------------------------------------|----------------------|----------------------------------------|
| /app                     | T-SHELL  | import-step<br>kpi-cards,<br>aging-cards,<br>alerts,<br>charts | acct.dashboard.*     | GET /api/accounting/dashboard          |
| /app/accounts            | T-INDEX  | tree-table,<br>filters,<br>create-drawer                       | accounts.*           | GET /api/accounts                      |
| /app/journals            | T-INDEX  | dense-table,<br>filters,<br>period-badge                       | journals.*           | GET /api/journals                      |
| /app/journals/new        | T-DETAIL | line-grid,<br>totals-footer,<br>attachment-uploader            | journal.draft.*      | GET/POST /api/journal-entries          |
| /app/journals/:id/review | T-DETAIL | review-panel,<br>warnings,<br>confirm-bar                      | journal.review.*     | GET/POST /api/journal-entries/:id/post |
| /app/sales-invoices/new  | T-DETAIL | invoice-form,<br>lines-grid,<br>totals-rail                    | salesInvoice.draft.* | POST /api/sales-invoices               |
| /app/purchase-bills/new  | T-DETAIL | bill-form,<br>account-picker,                                  | purchaseBill.draft.* | POST /api/purchase-bills               |

| route               | template | primary components                            | sample data keys    | mock API endpoints           |
|---------------------|----------|-----------------------------------------------|---------------------|------------------------------|
|                     |          | attachments                                   |                     |                              |
| /app/payments/new   | T-DETAIL | payment-form, allocation-grid, summary-card   | payment.draft.*     | POST /api/payments           |
| /app/reconciliation | T-RECON  | statement-list, suggestion-panel, action-tray | recon.*             | GET /api/reconciliation/*    |
| /app/tax/settings   | T-DETAIL | tax-code-table, rate-form, effective-dates    | tax.settings.*      | GET/PUT /api/tax/settings    |
| /app/tax/returns    | T-INDEX  | return-table, status-badges, filing-drawer    | tax.returns.*       | GET /api/tax/returns         |
| /app/reports/*      | T-REPORT | report-table, chart, export-bar, date-range   | reports.financial.* | GET /api/reports/*           |
| /app/period-close   | T-DETAIL | checklist, blocker-list, confirm-modal        | periodClose.*       | GET/POST /api/period-close/* |
| /app/imports        | T-       | job-                                          | imports.*           | GET/POST                     |

| route           | template | primary components                           | sample data keys      | mock API endpoints      |
|-----------------|----------|----------------------------------------------|-----------------------|-------------------------|
|                 | INDEX    | table, mapping-grid, diff-drawer             |                       | /api/imports/*          |
| /app/settings/* | T-DETAIL | company-form, users-table, localization-form | settings.accounting.* | GET/PUT /api/settings/* |
| /admin/*        | T-ADMIN  | tenant-table, pack-timeline, audit-log       | admin.accounting.*    | GET /api/admin/*        |

## Open questions and limitations

Primary country, tax regime, payment rails, bank-feed scope, localization depth, video relevance, and AI vendor choice all remain **unspecified**. Those are not minor details: they determine whether the accounting manifest stops at one-country VAT/GST support or grows into a much larger ERP-style program, and they determine whether optional media surfaces stay optional or become a cost driver. The free-only plan above is realistic for UI production and MVP integration, but official pricing docs show that analytics volume, object storage reads, email volume, and especially video delivery can outgrow no-cost tiers quickly. [16]

---

[1] Web Content Accessibility Guidelines (WCAG) 2.2

[https://www.w3.org/TR/WCAG22/?utm\\_source=chatgpt.com](https://www.w3.org/TR/WCAG22/?utm_source=chatgpt.com)

[2] Next.js Docs: App Router

[https://nextjs.org/docs/app?utm\\_source=chatgpt.com](https://nextjs.org/docs/app?utm_source=chatgpt.com)

[3] <https://authjs.dev/getting-started>

<https://authjs.dev/getting-started>

[4] Introduction - Shadcn UI

[https://ui.shadcn.com/docs?utm\\_source=chatgpt.com](https://ui.shadcn.com/docs?utm_source=chatgpt.com)



[5] <https://tiptap.dev/docs/editor/getting-started/overview>

<https://tiptap.dev/docs/editor/getting-started/overview>

[6] <https://akaunting.com/hc/docs/double-entry-accounting/>

<https://akaunting.com/hc/docs/double-entry-accounting/>

[7] <https://developers.cloudflare.com/pages/llms-full.txt>

<https://developers.cloudflare.com/pages/llms-full.txt>

[8] Vercel Hobby Plan

[https://vercel.com/docs/plans/hobby?utm\\_source=chatgpt.com](https://vercel.com/docs/plans/hobby?utm_source=chatgpt.com)

[9] Neon pricing

[https://neon.com/pricing?utm\\_source=chatgpt.com](https://neon.com/pricing?utm_source=chatgpt.com)

[10] [16] <https://supabase.com/pricing>

<https://supabase.com/pricing>

[11] <https://developers.cloudflare.com/r2/pricing/>

<https://developers.cloudflare.com/r2/pricing/>

[12] <https://github.com/seaweedfs/seaweedfs>

<https://github.com/seaweedfs/seaweedfs>

[13] PostHog – We make dev tools for product engineers

[https://posthog.com/?utm\\_source=chatgpt.com](https://posthog.com/?utm_source=chatgpt.com)

[14] <https://resend.com/pricing>

<https://resend.com/pricing>

[15] <https://developers.cloudflare.com/stream/pricing/>

<https://developers.cloudflare.com/stream/pricing/>